Ans. (d) : From question,

$$\frac{91\times92\times93\times94\times95\times96\times97\times98\times99}{13\times97}$$

Remainder = 0

59. Consider the following statements in respect of a rectangular sheet of length 20 cm and breadth 8 cm:

1. It is possible to cut the sheet exactly into 4 square sheets.

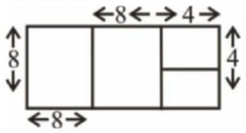
2. It is possible to cut the sheet into 10 triangular sheets of equal area.

Which of the above statements is/are correct?

(a) 1 only (b) 2 only

(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (c) : From statement 1



It is possible to construct the four square.

Hence, statement 1 is correct.

From statement 2

$$10\times\frac{1}{2}\times8\times a = 20\times8$$

$$a = 4$$

Hence, both statement 1 and 2 are correct.

60. When 70% of a number x is added to another number y, the sum becomes 165% of the value of y. When 60% of the number x is added to another number z, then the sum becomes 165% of the value of z. Which one of the following is correct?

(a) $z < x < y$ (b) $x < y < z$

(c) $y < x < z$ (d) $z < y < x$

Ans. (a) : According to the question,

$$x\times\frac{70}{100}+y = y\times\frac{165}{100}$$

$$x\times\frac{70}{100} = y\times\frac{65}{100}$$

$$\frac{x}{y} = \frac{13}{14}$$

Similarly, $x\times\frac{60}{100}+z = z\times\frac{65}{100}$

$$x\times\frac{60}{100} = z\times\frac{65}{100} \Rightarrow \frac{x}{z} = \frac{13}{12}$$

$x : y : z = 13 : 14 : 12$

$\therefore z < x < y$

Directions for the following 3 (three) items:

Read the following two passages and answer the items that follow the passages. Your answers to these items should be based on the passages only.

Passage–1

The majority of people who fail to accumulate money sufficient for their needs, are generally, easily influenced by the opinions of others. They permit the newspapers and the gossiping neighbors to do their thinking for them. Opinions are the cheapest commodities on the earth. Everyone has a flock of opinions ready them. If you are influenced by opinions when you reach decisions, you will not succeed in any undertaking.

61. Which one of the following is implied by the passage?

(a) Most of the people do not accumulate money for their needs.

(b) Most of the people never fail to accumulate money for their needs.

(c) There are people who fail to accumulate money for their needs.

(d) There is no need to accumulate money.

Ans. (c) : There are people who fail to a accumulate money for their needs is implied by the passage.

62. What is the main idea of the passage?

(a) People should not be influenced by the opinions of others.

(b) People should accumulate as much money as they can.

(c) People should neither give nor accept the opinions.

(d) People will succeed in any undertaking if they do not accept any opinion at all.

Ans. (a) : The main idea of this passage is that the people should not be influenced by the opinions of others.

Passage -2

" The social order is a sacred right which is the basis of all other rights. Nevertheless, this right does not come from nature, and must therefore be founded on conventions."

63. With reference to the above passage, which of the following statements is/are correct?

1. Conventions are the sources of rights of man.
2. Rights of man can be exercised only when there is a social order.

Select the correct answer using the code given below.

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

Ans. (c) : With reference to the above passage, both of the statements are correct. The statements are–

1. Conventions are the sources of rights of man.
2. Rights of man can be exercised only when there is a social order.

64. Two candidates X and Y contested an election. 80% of voters cast their vote and there were no invalid votes. There was no NOTA (None of the above) option. X got 56% of the votes cast and won by 1440 votes. What is the total number of voters in the voters list?

(a) 15000 (b) 12000

(c) 9600 (d) 5000

Ans. (a) :

Let the number of voters in the voter list = x

Voting people = 80x/100

Percentage difference of votes obtained by X and Y

$$1440 = \frac{80x}{100}\times\frac{(56-44)}{100}$$

$$1440 = \frac{8x}{10}\times\frac{12}{100}$$

$x = 15000$

65. What is the smallest number greater than 1000 that when divided by any one of the numbers 6, 9, 12, 15, 18, leaves a remainder of 3?

(a) 1063 (b) 1037

(c) 1083 (d) 1183

Ans. (c) : LCM of 6, 9, 12, 15 18 = 180

Required number = 180k + 3

K = 6 taking

= 1080 + 3 = 1083

66. Let p be a two-digit number and q be the number consisting of same digits written in reverse order. If p × q = 2430, then what is the difference between p and q?

(a) 45

(b) 27

(c) 18

(d) 9

Ans. (d) : From question,

$p \times q = 2430$

$p \times q = 54 \times 45$

$\therefore p = 54, q = 45$

$p - q = 9$

67. Consider the following statements in respect of two natural numbers p and q such that p is a prime number and q is a composite umber:

1. $p \times q$ can be an add number.
2. q/p can be a prime number.
3. $p + q$ can be a prime number.

Which of the above statements are correct?

(a) 1 and 2 only (b) 2 and 3 only

(c) 1 and 3 only (d) 1, 2 and 3

Ans. (d) : Let, p = 3, q = 9

From statement 1

$p\times q = 27$ (Odd number)

From statement 2

$$\therefore \frac{21}{3} = 7$$

So, statement 2 is correct.

If p = 3, q = 4

$\therefore p + q = 7$

Hence, statement 3 is correct.

68. Consider the following statements:

1. Between 3:16 p.m. and 3:17 p.m. both hour hand and minute hand coincide.
2. Between 4:58 p.m. and 4:59 p.m. both minute hand and second hand coincide.

Which of the above statements is/are correct?

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

Ans. (c) : From statement 1

$$\theta = \frac{11M}{2} \sim 30H$$

$$0 = \frac{11M}{2} \sim 30\times3$$

$$\frac{11M}{2} = 90$$

$$M = \frac{180}{11}$$

$$M = 16\frac{4}{11}$$

Minute hand and hour hand will meet at 3: 16 $\frac{4}{11}$ O'clock.

Both statements 1 and 2 are correct.

69. There are two containers X and Y. X contains 100 ml of milk and Y contains 100 ml of water, 20 ml of milk from X is transferred to Y. After mixing well, 20 ml of the mixture in Y is transferred back to X. If m denotes the proportion of milk in X and n denotes the proportion of water in Y, then which on of the following is correct?